Roll No.....

End Semester Examination

M. Tech: (Biomedical Engineering) I[st] Semester

Biomedical Electronics- BEM 103

Time: 3.00 Hrs

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1	(20 marks)	
(a)	Explain the electrical and mechanical activities of the heart and how they relate to ECG signals.	CO1
(b)	Describe the structure and function of the respiratory system and how respiration rate can be measured.	CO1
(c)	Discuss the physiology of hearing and the role of the auditory system in sound signal transduction	CO1
Q2	(20 marks)	
(a)	Explain the origin and characteristics of bioelectric signals in excitable tissues such as nerves and muscles.	CO2
(b)	Explain how EEG signals reflect brain activity. Discuss the different types of brain waves and their physiological significance.	CO2
(c)	Describe the process of recording electromyogram (EMG) signals and explain how EMG is used to study muscle function.	CO2
Q3	(20 marks)	
(a)	Describe different types of electrodes used in ECG, EEG, and EMG. Explain how each is designed to suit its application.	CO3
(b)	Discuss the construction, working, and biomedical uses of a Linear Variable Differential Transformer (LVDT).	CO3
(c)	Explain the principle and operation of a capacitive pressure transducer with a neat diagram.	CO3
Q4	(20 marks)	
(a)	Explain the working of an instrumentation amplifier with a neat circuit diagram. Derive the expression for its voltage gain and list its advantages in biomedical applications.	CO4
(b)	Describe the complete recording system for EMG. Include the types of electrodes used, amplifiers, filters, and display devices. Discuss the main challenges in EMG signal acquisition.	CO4
(c)	Explain the importance of signal conditioning in biomedical instrumentation. Discuss how amplifiers and filters improve the signal quality in ECG or EEG recording systems.	CO4
Q5	(20 marks)	
(a)	What are the biological effects of X-rays? Explain the safety measures adopted in medical X-ray imaging systems.	CO5
(b)	Discuss the construction and working of an infrared thermography system. What are its clinical applications and limitations?	CO5
(c)	Explain the principle and working of a cardiac defibrillator. Describe different types such as AC, DC, and implantable defibrillators.	CO6